Archita Lahiri

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Google Scholar

Research Interests

Machine Learning • Deep Learning • Artificial Intelligence • Mathematics • Data Analytics • Software Development

Education

Rajshahi University of Engineering & Technology **Session 2025-2026**
Bachelor of Science in Computer Science and Engineering
CGPA: **3.71 / 4.00**

Naogaon Govt. College, Naogaon **Session 2020-2021**
Higher Secondary Certificate (HSC), Science
GPA: **5.00 / 5.00**

Naogaon Govt. Girls High School **Session 2018-2019**
Secondary School Certificate (SSC), Science
GPA: **5.00 / 5.00**

Publications

Impact of Stain Normalization and Background Filtering on Deep Learning & Transformer-based Models for Ovarian Cancer Histopathology Classification *ICCIT 2025*

- Conducted a controlled study across architectures and preprocessing pipelines on the UBC-OCEAN dataset.
- Demonstrated that background filtering consistently improves classification performance across architectures and identified the best-performing pipeline configuration.

LunaXNet: A Dual-Path Hyperspectral AI Framework for Lunar Mineral Mapping with Explainability and Uncertainty Estimation *On going*

- Developed a deep-learning framework for lunar mineral mapping from Chandrayaan-1 M^3 hyperspectral data, comparing classical and deep approaches in a shared latent space.
- Contributed to methodology, software, and data curation, including explainability and uncertainty estimation.

RoboSight++: Robust Object Detection *Ongoing*

- Working on robust and explainable object detection methods to improve the reliability of computer vision systems in real-world robotic environments.

Undergraduate Thesis

Reinforcement Learning-Driven Adversarial Evasion Attack on Machine Learning-Based Network Intrusion Detection Systems

Supervisor: Md. Azmain Yakin Srizon, RUET

- Developed PPO- and A2C-based adaptive adversarial attacks against ML-based NIDS and evaluated their transferability across LR, DT, RF, XGBoost, and MLP models using CIC-IDS2017 and LycoS-UNICAS-IDS2018.

Experience

Industrial Attachment **1 Month**
SELISE Digital Platforms, Bangladesh

- Gained hands-on experience with professional software development processes and industry-standard workflows.
- Practiced **Scrum methodology** through regular team meetings, task planning, and daily development activities.
- Gained initial hands-on experience with **Microsoft Azure** for application deployment and cloud-based services.
- Worked with different **deployment tools and techniques** used in modern software development environments.
- Strengthened practical skills in teamwork, communication, problem-solving, version control, and collaborative development.

Projects

Shohochori: A Platform for Elderly Users *React, Node.js, Spring Boot, Azure, Git*

- Developed a web-based platform designed to help elderly users access essential digital services through a simple and user-friendly interface.
- Implemented two versions of the application: one using **React and Node.js** and another using **React and Java Spring Boot**.
- Explored a **microservices-based architecture** to separate different application functionalities and improve modularity.
- Deployed the React and Node.js version on **Microsoft Azure** and used **Git** for version control and source-code management.

Job Portal Android Application *Kotlin, Firebase*

- Developed an Android-based job portal application using **Kotlin**, allowing users to browse and manage job-related information.

- Integrated **Firebase** for backend services, database management, and user authentication.

Routine Management System *Java*

- Developed a routine management application in **Java** as an academic project for an Object-Oriented Programming course.
- Applied core OOP concepts such as classes, objects, inheritance, encapsulation, and modular program design.

Web Application *Laravel, MySQL*

- Developed a database-driven web application using the **Laravel** PHP framework.
- Used **MySQL** for database design, storage, and management of application data.

Certifications

- **Young Learner's Research Lab** – Researcher,2024-2026
- **EDGE Project Certification** – Successfully completed a course under the EDGE Project.
- **Supervised Machine Learning: Regression and Classification** – Coursera.
- **National Math Olympiad** – Participated in the National Round.
- **NGPC 2022** – Competed in the NGPC 2022 National Round.

Honors and Awards

- 2nd place , Nascenia - AI Hackathon on Kaggle, 2026
- Finalist, Innovation World Cup(Indonesia) 2025
- Finalist, Therap Javafest 2023,2024.
- Scholarship recipient based on SSC result 2018
- Academic scholarship recipient based on 1st, 2nd, and 3rd-year university results. 2023,2024,2025
- Participant, Regional Physics Olympiad. 2016, 2017
- Runners-Up, National Math Olympiad at school and college levels. 2018, 2019

Non-Academic Experience

- Member, RUET Competitive Programming Society 2023–2026
- Member, RUET RAPL 2022–2026
- Member, RUET Ovijatrik 2024–2026
- Member, Onuronon 2022–2026

Volunteering

Volunteer, RUET Ovijatrik Marathon

- Volunteered in a marathon organized by RUET Ovijatrik Club.
- Gained practical experience in event management, teamwork, and coordination.

Skills

- **Programming Languages:** C, C++, Python, Java, JavaScript, MySQL
- **Machine Learning & AI:** Machine Learning, Deep Learning, Artificial Intelligence
- **Technical Skills:** Data Analytics, Documentation, Technical Support
- **Soft Skills:** Presentation, Critical Thinking, Public Speaking, Communication, Teamwork

References

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